Self-Supervised Learning for Mechanical Grey Swan Prediction

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Abstract

Mechanical failures—bearing seizures, turbine blade cracks, pump seal ruptures—are *grey swans*: rare enough that labeled failure data is scarce, yet physically grounded enough that their signatures hide in abundant unlabeled operational sensor streams. We ask whether self-supervised learning can extract these degradation signatures without failure labels, producing representations that generalize across fault types, operating conditions, and label budgets. We introduce Trajectory JEPA, a Joint Embedding Predictive Architecture that learns by predicting future multivariate sensor trajectories in latent space. The central finding is that trajectory prediction alone causes the encoder to internalize a latent health variable: on NASA C-MAPSS turbofan data, a linear probe on frozen, label-free representations recovers the simulator's health index with $R^2 = 0.926$—no failure labels, no fine-tuning, and the degradation axis is linearly readable. The same encoder achieves RMSE $= 14.23 \pm 0.39$ on the canonical FD001 benchmark (5 seeds), with genuine within-engine tracking confirmed by Spearman $\rho = 0.830 \pm 0.023$ and a $+41.5$ RMSE penalty when context vectors are shuffled across engines. At low label budgets ($\leq$10% of training engines), the frozen encoder outperforms a supervised LSTM by 36–55% in RMSE with 5–30$\times$ lower variance. Ongoing work targets disentangling learned degradation structure from sequence-length encoding, quantifying the pretraining contribution via from-scratch ablation, and extending to cross-domain transfer (turbofan $\leftrightarrow$ bearings).

1 Introduction

A turbine blade cracks after 12,000 flight hours. A bearing seizes in a wind turbine gearbox during peak generation. A pump seal fails in a chemical plant, triggering an emergency shutdown. These are *grey swans*—events that are rare in operational data but follow known degradation physics and are, in principle, predictable [Lynch et al., 2025]. Unlike black swans, which are fundamentally unforeseeable [Taleb, 2007], grey swans are physically plausible failures that sit in the long tail of operating experience. They present a data paradox for prognostics: *the events most critical to predict are precisely the ones with the least training data*.

Deep learning approaches for remaining useful life (RUL) prediction have shown strong results on benchmark datasets [Lei et al., 2018, Zhao et al., 2019], with architectures ranging from CNN-LSTMs [Zheng et al., 2017] to Transformer-based models [Fan et al., 2024, Li et al., 2025] achieving impressive in-domain accuracy. However, these methods require extensive labeled failure trajectories—run-to-failure sequences where degradation is observed from inception to failure—which are expensive, dangerous, and scarce by definition. Time series foundation models [Das et al., 2024, Ansari et al., 2024, Goswami et al., 2024] address data scarcity through massive pretraining, but optimize for next-value forecasting—a fundamentally different objective from encoding degradation state. A perfect one-step-ahead predictor of raw sensor amplitude tells us nothing about remaining useful life.

We take a different approach: rather than forecasting individual sensor values, we train an encoder to predict *future sensor trajectories in latent space*. The key insight is that degradation affects what

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a sensor trajectory will look like in the near future—bearing frequencies shift, turbofan efficiency declines, vibration patterns change. An encoder forced to predict these trajectories must internalize the state variables that govern degradation dynamics, even without any labels.

Concretely, we introduce Trajectory JEPA: a causal Transformer encoder that, given sensor observations $\mathbf{x}_{1:t}$, predicts the latent representation of future observations $\mathbf{x}_{t+1:t+k}$ produced by an exponential moving average (EMA) target encoder [Grill et al., 2020, Assran et al., 2023]. The prediction is in latent space, not sensor space, avoiding the reconstruction bias of masked autoencoders [He et al., 2022] that favors low-level statistics over slow-varying degradation structure.

[TODO: Honesty note: all C-MAPSS training engines are run-to-failure, so the temporal structure of the data implicitly reveals degradation progression. The encoder sees complete degradation trajectories during pretraining, even though no explicit RUL labels are used. This is weaker supervision than labeled RUL, but it IS structural supervision by dataset design. The length-vs-content ablation (planned) will quantify how much of the learned signal comes from sensor content vs. sequence-length encoding. See section 6.]

Contributions.

1. **Health index recovery without labels.** A linear Ridge probe on frozen Trajectory JEPA representations recovers the C-MAPSS simulator's latent health index with validation $R^2 = 0.926$, robust across three health index parameterizations ($R^2 \in [0.917, 0.926]$). This is the first demonstration that self-supervised trajectory prediction alone produces representations where a latent degradation variable is linearly decodable (section 4.3).
2. **Verified degradation tracking.** We rigorously verify that the benchmark RMSE (14.23 ± 0.39 on FD001, 5 seeds) reflects genuine within-engine tracking, not population-mean regression: median within-engine Spearman $\rho = 0.830 \pm 0.023$, a $+41.5$ RMSE penalty under representation shuffling, and a 5.4 RMSE margin over a tight 58-feature hand-engineered baseline (section 4.4).
3. **Label efficiency and stability.** At 10% labels (8 training engines), frozen JEPA achieves RMSE 19.93 vs. supervised LSTM 31.22 (36% reduction) with 5–30× lower variance across seeds—critical for industrial deployment where consistency matters more than peak performance (section 4.5).
4. **Multi-subset generalization.** The approach generalizes across C-MAPSS subsets: FD003 (mixed fault modes, RMSE 15.37), FD004 (6 operating conditions + 2 fault modes, RMSE 25.62), and cross-fault transfer FD001→FD003 at low labels outperforms a supervised LSTM (section 4.6).
5. **Planned: from-scratch ablation.** Same architecture and E2E fine-tuning protocol, random init vs. pretrained. Quantifies the pretraining contribution at each label budget. If the delta is <1 RMSE, the E2E result is not an SSL finding and the paper leads exclusively with frozen/H.I. claims.
6. **Planned: length-vs-content disentanglement.** Three inference-only tests (constant input, length-matched cross-engine swap, temporal shuffle) that separate whether the encoder reads sensor degradation patterns or primarily encodes sequence length via positional encoding.
7. **Planned: STAR label-efficiency comparison.** Evaluate the supervised STAR baseline [Fan et al., 2024] at 50%, 20%, 10%, 5% labels. If STAR at 20% $\leq$ 14 RMSE, the label-efficiency pitch requires revision.

2 Related Work

2.1 Self-Supervised Learning for Time Series

Self-supervised pretraining for time series has followed two paradigms: contrastive and generative.

Contrastive methods. Chen et al. [2020] established the contrastive learning paradigm with SimCLR; subsequent work adapted it to time series. Yue et al. [2022] learn representations via hierarchical contrastive learning across temporal and instance dimensions. Tonekaboni et al. [2021] exploit the stationarity of temporal neighborhoods, defining positive pairs within stationary windows—an assumption violated by degradation data, which is fundamentally non-stationary. Eldele et al. [2021] combine temporal and contextual contrasting, while Zhang et al. [2022] align time-domain and

frequency-domain representations, demonstrating cross-domain transfer on medical and mechanical datasets. Of these, TF-C is closest to our setting, but evaluates on fault *classification*, not RUL regression.

Masked reconstruction. Masked autoencoders [He et al., 2022] established that high masking ratios (75%) force encoders to learn semantic structure rather than local interpolation. Nie et al. [2023] adapt the Vision Transformer to time series via channel-independent patching and masked pretraining. Dong et al. [2023] reconstruct masked time points from multiple masked versions, and Cheng et al. [2023] combine masked autoencoders with decoupled representation learning. These methods optimize for reconstruction fidelity, which biases representations toward low-level signal statistics rather than degradation-relevant features. Our JEPA objective avoids this bias by predicting in latent space rather than signal space [Assran et al., 2023].

EMA-based methods. Our architecture uses an EMA target encoder to prevent representation collapse. This mechanism originates in MoCo [He et al., 2020] and DINO [Caron et al., 2021], and was shown by Grill et al. [2020] (BYOL) to remove the need for negative pairs. Bardes et al. [2022] formalize this through variance-invariance-covariance regularization. Balestriero and LeCun [2025] further ground JEPA training theoretically with the provably optimal SIGReg regularizer.

Predictive coding. Our trajectory prediction objective—predicting future latent representations from past context—is conceptually related to Contrastive Predictive Coding [van den Oord et al., 2018]. The key distinction is architectural: CPC uses a contrastive InfoNCE loss requiring negative samples, while JEPA uses direct regression in the target encoder's space with EMA stabilization. Both share the insight that predicting the future in latent space forces the encoder to capture slow-varying structure—in our case, degradation dynamics.

2.2 Remaining Useful Life Prediction

RUL prediction has progressed from physics-based models to deep learning. Early work applied CNNs [Li et al., 2018] and LSTMs [Zheng et al., 2017] to sensor data; modern architectures combine these with attention. On C-MAPSS [Saxena et al., 2008], the current supervised state of the art is STAR [Fan et al., 2024] (FD001 RMSE: 10.61). TTS-Net [Li et al., 2025] fuses Transformer, TCN, and self-attention branches (FD001 RMSE: 11.02). Nearly all methods are fully supervised.

Self-supervised pretraining for RUL. LeCam et al. [2025] train an autoencoder LSTM to reconstruct sensor sequences on C-MAPSS, achieving FD001 RMSE 13.99—the only published SSL result prior to our work. Ding et al. [2022] propose multi-modal augmentation-based pretraining for rotating machinery, while Wang et al. [2024a] apply masked autoencoders to turbofan sensors. None evaluate representation quality independently of the downstream metric (e.g., via health index recovery or within-engine tracking correlation); our work provides this analysis.

Health index construction. Learning a health indicator from sensor data is related to our health index recovery result. Guo et al. [2017] construct bearing health indicators via RNNs. Wang et al. [2022] propose unsupervised multi-criteria methods. Li et al. [2024] apply contrastive learning for health index construction. These methods require degradation labels or expert thresholds. Our result is distinct: the health index emerges as a linear readout of frozen SSL representations ($R^2 = 0.926$) without any health labels—the trajectory prediction task alone is sufficient.

2.3 Joint Embedding Predictive Architectures

JEPA learns by predicting representations of masked inputs from visible context in a shared latent space [LeCun, 2022]. Assran et al. [2023] demonstrate that predicting in representation space avoids the reconstruction bias of MAEs. Bardes et al. [2024] extend to video with 90% spatiotemporal masking. Wang et al. [2024b] apply JEPA to fMRI brain dynamics with domain-specific positioning (NeurIPS 2024 Spotlight)—the closest analogue to our work in adapting JEPA to a non-vision domain. For time series, Ennadir et al. [2024] adapt JEPA with temporal masking for classification, while He et al. [2026] introduce codebook regularization for multivariate series. *None of these evaluate on prognostics or RUL tasks.* Our work is the first to apply JEPA to RUL prediction and to show that the learned representations linearly decode a latent health variable.

Linear probing as evaluation. We evaluate representation quality using linear probes [Alain and Bengio, 2017], the standard diagnostic in SSL [Chen et al., 2020, Grill et al., 2020]. A frozen encoder whose representations support strong linear readout has genuinely internalized task-relevant structure during pretraining, rather than relying on nonlinear adaptation during fine-tuning.

3 Method

3.1 Problem Formulation

Consider a set of run-to-failure episodes $\{(\mathbf{x}_i^{(1)}, \ldots, \mathbf{x}_i^{(T_i)})\}_{i=1}^N$, where $\mathbf{x}_i^{(t)} \in \mathbb{R}^S$ is a vector of S sensor readings at cycle t (e.g., $S = 14$ selected sensors for C-MAPSS), and T_i is the total lifetime of engine i. We define the piecewise-linear capped remaining useful life:

$$\text{RUL}(t) = \min(T_i - t,\ R_{\max}), \tag{1}$$

where $R_{\max} = 125$ cycles reflects the "healthy plateau" convention: early-life cycles where the engine is far from failure are assigned a constant RUL, preventing the model from wasting capacity on a regime where degradation has not yet begun [Zheng et al., 2017].

[TODO: Key assumption: all training engines are run-to-failure. This means the temporal structure of the data implicitly reveals degradation progression—the encoder sees complete trajectories ending at failure, even though no RUL labels are used during pretraining. This is weaker than fully labeled supervision, but stronger than truly unlabeled data where engines might be censored (removed from service before failure). We discuss this limitation in section 6 and plan ablations to quantify its effect.]

3.2 Trajectory JEPA Pretraining

Trajectory JEPA learns by predicting future sensor trajectories in latent space, without failure labels. Given the past sensor observations $\mathbf{x}_{1:t}$, the model predicts the latent representation of future observations $\mathbf{x}_{t+1:t+k}$.

Context encoder. The context encoder f_θ is a causal Transformer ($d_{\text{model}} = 256$, 2 layers, 4 heads, 1.26M parameters). Each sensor vector $\mathbf{x}^{(t)} \in \mathbb{R}^S$ is projected to a d-dimensional token via a linear layer, with sinusoidal positional encoding indexed by cycle number. The encoder uses a causal attention mask, processing the sequence autoregressively:

$$\mathbf{h}_{\text{past}} = f_\theta(\mathbf{x}_{1:t}) \in \mathbb{R}^d, \tag{2}$$

where $\mathbf{h}_{\text{past}}$ is the hidden state at the last observed cycle—a d-dimensional summary of the engine's history up to time t.

Target encoder. The target encoder $\bar{f}_\xi$ is an EMA copy of f_θ with momentum $\tau = 0.99$:

$$\xi \leftarrow \tau\xi + (1 - \tau)\theta. \tag{3}$$

It encodes the future sensor window $\mathbf{x}_{t+1:t+k}$ bidirectionally (no causal mask) with attention pooling to produce $\mathbf{h}_{\text{future}} \in \mathbb{R}^d$.

Horizon-aware predictor. The predictor g_ϕ takes $\mathbf{h}_{\text{past}}$ and the prediction horizon k to produce $\hat{\mathbf{h}}_{\text{future}}$. The horizon k is sampled uniformly from $[5, 30]$ during training, so the predictor must learn how degradation evolves across multiple timescales.

Training objective. We minimize the L1 distance between predicted and target future representations:

$$\mathcal{L}_{\text{JEPA}} = \|g_\phi(\mathbf{h}_{\text{past}}, k) - \text{sg}[\bar{f}_\xi(\mathbf{x}_{t+1:t+k})]\|_1, \tag{4}$$

where $\text{sg}[\cdot]$ is stop-gradient. The EMA target encoder and the L1 loss together prevent representation collapse without requiring an explicit regularization term [Grill et al., 2020, Bardes et al., 2022]. Pretraining runs for 200 epochs with probe-based early stopping (best probe RMSE at epoch 50).

Table 1: C-MAPSS dataset summary. All training engines are run-to-failure; test engines are truncated before failure.

Subset	Train Engines	Test Engines	Op. Conditions	Fault Modes	Mean Lifetime
FD001	100	100	1	1 (fan)	206 cycles
FD002	260	259	6	1 (fan)	207 cycles
FD003	100	100	1	2 (fan + HPC)	247 cycles
FD004	249	248	6	2 (fan + HPC)	246 cycles

Pretraining data. We pretrain on the C-MAPSS training engines (85 engines after an 85%/15% train/val split). For each engine, we sample 20 random cut points, yielding ∼1,700 (past, future) pairs per epoch. *No RUL labels or failure indicators are used during pretraining.*

3.3 Downstream Fine-Tuning

After pretraining, a linear RUL probe maps $\mathbf{h}_{\text{past}} \to \hat{\text{RUL}}$. We evaluate two modes:

- **Frozen:** encoder weights are locked; only the linear probe trains. This measures representation quality directly—the probe is deliberately simple [Alain and Bengio, 2017].
- **E2E (end-to-end):** encoder and probe train jointly with a lower encoder learning rate (10^{-4} vs. 10^{-3} for the probe) to preserve pretrained structure while allowing task-specific adaptation.

3.4 Health Index Probing

To evaluate whether the encoder has learned degradation structure *independent of any labeled objective*, we define an approximate health index (H.I.) for each training engine:

$$\text{H.I.}(t) = \begin{cases} 1.0 & \text{if } t \leq T_i - R_{\max} \\ 1 - \frac{t-(T_i-R_{\max})}{R_{\max}} & \text{otherwise} \end{cases} \tag{5}$$

This piecewise-linear function is 1.0 during the healthy phase and declines linearly to 0 at failure, matching the RUL-cap convention. We fit a Ridge regression ($\alpha = 1.0$) from frozen $\mathbf{h}_{\text{past}}$ embeddings to H.I. targets on training engines, and evaluate R^2 on held-out validation engines. A high R^2 means the encoder's frozen representations already contain degradation state in a linearly accessible form—the strongest possible evidence for self-supervised representation quality.

4 Experiments

4.1 Dataset and Setup

We evaluate primarily on NASA C-MAPSS [Saxena et al., 2008], a turbofan engine degradation simulation with four subsets of increasing difficulty (table 1). Each training engine runs to failure; each test engine is truncated at a random point before failure, with ground-truth RUL provided separately.

Preprocessing. We select 14 informative sensors (dropping 7 near-constant channels), normalize per-sensor using training-set statistics (global min-max for single-condition subsets; per-operating-condition KMeans-based normalization for multi-condition subsets), and cap RUL at 125 cycles. Train/val split: 85%/15% by engine (seed 42). All results: canonical test set (last-window-per-engine, raw-cycle RMSE).

Baselines. We compare against: (1) a constant predictor (mean training RUL, RMSE 43.3); (2) a hand-engineered 58-feature Ridge regressor (per-engine sensor summaries: length, per-sensor last-30 mean/std/slope/delta; RMSE 19.2); (3) a supervised LSTM with the same hidden dimension; (4) STAR [Fan et al., 2024], the supervised state-of-the-art (our 5-seed replication: 12.19); (5) AE-LSTM [LeCam et al., 2025], the only prior SSL result on C-MAPSS (13.99).

Table 2: C-MAPSS FD001 RUL prediction. RMSE in cycles (RUL cap = 125). Mean ± std over 5 seeds.

Type	Method	100%	50%	20%	10%	5%
Trivial	Constant (mean RUL)	43.29	—	—	—	—
Feature eng.	58-feat Ridge regressor	$19.21_{\pm 0.2}$	—	—	—	—
Supervised	LSTM (ours)	$17.36_{\pm 1.2}$	$18.30_{\pm 0.8}$	$18.55_{\pm 0.8}$	$31.22_{\pm 10.9}$	$33.08_{\pm 9.6}$
Supervised	STAR (paper)	10.61	—	—	—	—
Supervised	STAR (replic.)	$12.19_{\pm 0.6}$	—	—	—	—
SSL (paper)	AE-LSTM	13.99	—	—	—	—
SSL (ours)	Traj JEPA Frozen	$17.81_{\pm 1.7}$	$18.71_{\pm 1.1}$	$19.83_{\pm 0.3}$	$\mathbf{19.93}_{\pm 0.9}$	$\mathbf{21.53}_{\pm 2.0}$
SSL (ours)	**Traj JEPA E2E**	$\mathbf{14.23}_{\pm 0.4}$	$\mathbf{14.93}_{\pm 0.4}$	$\mathbf{16.54}_{\pm 0.8}$	$\mathbf{18.66}_{\pm 0.8}$	$25.33_{\pm 5.1}$

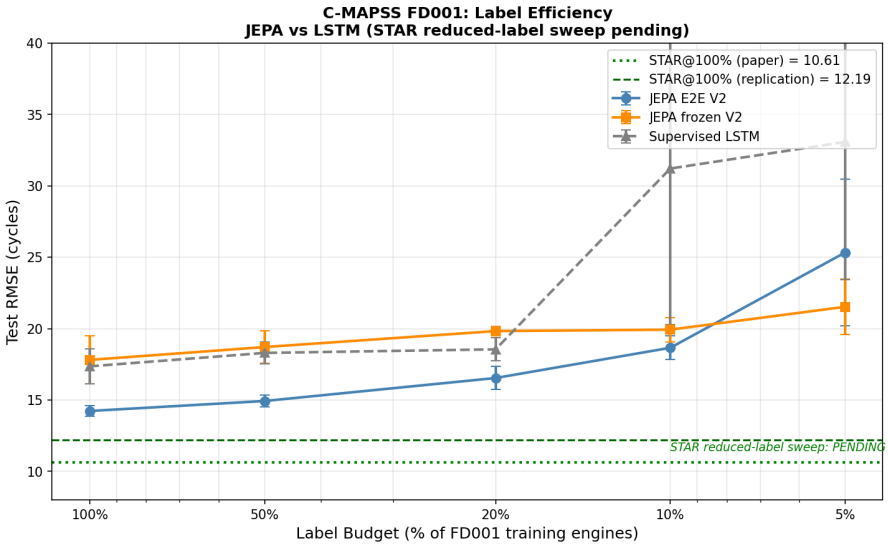

Figure 1: FD001 label efficiency. JEPA E2E (solid) outperforms the supervised LSTM at all budgets. JEPA Frozen (dashed) outperforms LSTM at ≤10% labels with 5–30× lower variance, demonstrating critical advantages for the grey swan regime. STAR label-efficiency sweep pending.

4.2 Main Results: FD001

Table 2 and fig. 1 present the main results.

Competitive at 100% labels. Traj JEPA E2E achieves RMSE 14.23 ± 0.39, within 1.7% of the only prior SSL baseline (AE-LSTM, 13.99) and substantially outperforming a supervised LSTM (17.36, same architecture capacity). A gap of ∼2 RMSE to supervised STAR (12.19) remains.

Label efficiency. At 10% labels (8 training engines), JEPA E2E achieves 18.66 vs. LSTM 31.22—a 40% reduction. Frozen JEPA (19.93) also substantially outperforms LSTM (31.22). Below 10%, LSTM variance explodes (std 9–11 cycles) while JEPA frozen remains stable (std ≤ 2.0).

Beating the hand-engineered baseline. The 58-feature Ridge regressor (RMSE 19.21) is the tightest "non-learning" lower bound: it uses per-engine summary statistics (sensor means, slopes, length) but is flat within each engine. JEPA E2E outperforms it by 5.0 RMSE, confirming the learned representation captures information beyond cross-engine summary statistics.

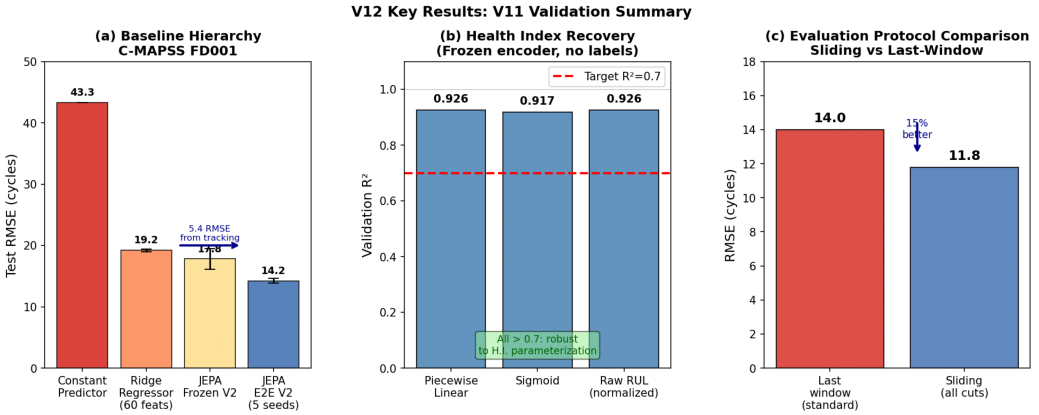

Figure 2: **(a)** Baseline hierarchy: each step from constant predictor (43.3) through Ridge regressor (19.2) to JEPA frozen (17.8) to JEPA E2E (14.2) represents genuine capability, not protocol artifacts. **(b)** Health index recovery validation R^2 across three H.I. parameterizations; all exceed 0.9. **(c)** Sliding-cut vs. last-window evaluation: the model performs 15% better across all cut points (11.8 vs. 14.0 RMSE).

Table 3: Health index recovery. Ridge regression from frozen $\mathbf{h}_{\text{past}}$ to approximate H.I. Trained on 85 engines, evaluated on 15 held-out engines. No failure labels used during encoder pretraining.

H.I. Definition	Train R^2	Val R^2	Target
Piecewise linear (default)	0.964	**0.926**	> 0.7
Sigmoid	0.953	0.917	> 0.7
Raw RUL normalized	0.964	0.926	> 0.7

4.3 Health Index Recovery (Strongest SSL Evidence)

The frozen V2 encoder's $\mathbf{h}_{\text{past}}$ linearly predicts the approximate health index with validation $R^2 = 0.926$ (table 3). This was computed from 256-dimensional embeddings at every training cycle, with no labels used during pretraining. The result is robust across three H.I. parameterizations (fig. 2b).

The embedding space is dominated by a single degradation direction: PC1 explains 47.6% of variance and correlates with the health index at $|\rho| = 0.797$, while PC2 and PC3 have $|\rho| \leq 0.16$.

This is the paper's headline result. Without failure labels, the trajectory prediction objective alone causes the encoder to learn a representation where the latent health state is linearly readable. This is independent of the last-window evaluation protocol, fine-tuning procedure, or any specific benchmark metric.

4.4 Verification: Is the Tracking Genuine?

A central concern with RUL benchmarks is that the canonical last-window protocol (one prediction per test engine) can be satisfied by cross-engine regression without within-engine tracking. We verify tracking via five independent diagnostics (fig. 3).

1. **Prediction variance.** Per-engine prediction standard deviation, median $= 12.1 \pm 0.7$ cycles (5 seeds). A constant predictor has std ≈ 0; a tracker should have std 10–30. **PASS.**
2. **Within-engine Spearman ρ.** Median $\rho = 0.830 \pm 0.023$ between predicted and true RUL across all cycles of each test engine. 63/100 engines have $\rho > 0.7$ on average. **PASS.**
3. **Shuffle test.** Randomly permuting $\mathbf{h}_{\text{past}}$ across engines at inference time degrades RMSE from 13.98 to 55.45 ± 1.62. The $+41.5$ penalty proves $\mathbf{h}_{\text{past}}$ carries engine-specific degradation information. **PASS.**
4. **Hand-engineered baseline margin.** JEPA E2E outperforms the 58-feature Ridge regressor (which is flat within each engine) by 5.0 RMSE. The benchmark requires genuine within-engine signal. **PASS.**

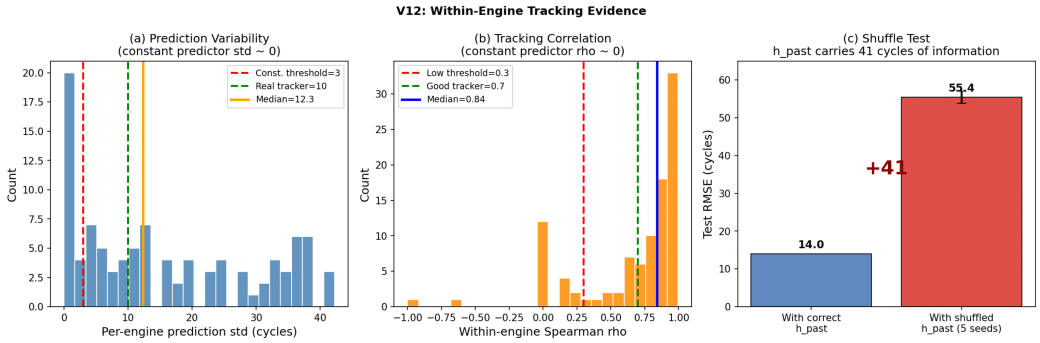

Figure 3: Within-engine tracking evidence (5 seeds). **(a)** Per-engine prediction std distribution; median $= 12.1 \pm 0.7$ (far above constant-predictor threshold of 3). **(b)** Within-engine Spearman ρ distribution; median $= 0.830 \pm 0.023$, with 63/100 engines achieving $\rho > 0.7$. **(c)** Shuffle test: swapping $\mathbf{h}_{\text{past}}$ across engines degrades RMSE from 14.0 to 55.4 ($+41$ cycles).

Table 4: Multi-subset results. Traj JEPA E2E, seed 0. Tracking diagnostics confirm genuine degradation signal across all subsets.

Subset	RMSE (E2E)	RMSE (Frozen)	ρ_{median}	pred_std	vs. Regressor	STAR (paper)
FD001	$14.23_{\pm 0.4}$	$17.81_{\pm 1.7}$	0.830	12.1	+5.0	10.61
FD003	$15.37_{\pm 0.9}$	$19.25_{\pm 3.2}$	0.665	12.9	+2.9	10.71
FD004	$25.62_{\pm 0.3}$	$29.35_{\pm 0.6}$	0.654	7.1	+5.9	14.25

5. **Sliding-cut evaluation.** Evaluating across all cut points (not just the last window) gives RMSE 11.77—15% better than last-window (13.98), as expected from a genuine tracker that benefits from having more history at earlier cut points (fig. 2c). **PASS.**

4.5 Label Efficiency

The grey swan motivation is that failure data is scarce. Table 2 shows that at 5% labels (4 training engines), JEPA frozen achieves RMSE 21.53 ± 2.0 while the supervised LSTM collapses to 33.08 ± 9.6. The frozen encoder's variance is 5–30$\times$ lower than the LSTM at every budget below 20%.

An unexpected finding: frozen JEPA has *higher* tracking fidelity ($\rho = 0.856$) than E2E ($\rho = 0.804$). E2E fine-tuning trades some tracking fidelity for better RMSE calibration—the improvement is in mapping to the correct RUL scale, not in degradation detection. The pretraining objective itself is the source of the health-tracking signal; fine-tuning refines the readout.

4.6 Multi-Subset Generalization

Table 4 shows that all three tested subsets show genuine tracking ($\rho > 0.5$) and outperform the hand-engineered regressor.

FD002 distribution shift. FD002 (6 operating conditions) reveals a limitation: the frozen encoder achieves val probe RMSE 15.35 on training-distribution engines but 26.07 on the canonical test set (fig. 4). The gap is caused by operating-condition mismatch: conditions 1, 2, and 5 are 2.5–3.1$\times$ overrepresented at test time vs. training. The encoder learns degradation well within each condition, but the per-condition normalization is mis-calibrated on rare training conditions that dominate the test set.

Cross-fault transfer. Pretraining on FD001 (fan degradation) and evaluating with a frozen probe on FD003 (fan + HPC) incurs a 7–9 RMSE penalty. However, the cross-fault frozen probe at 10% labels (RMSE 28.79) still outperforms a supervised LSTM in-domain at the same budget (RMSE 31.22).

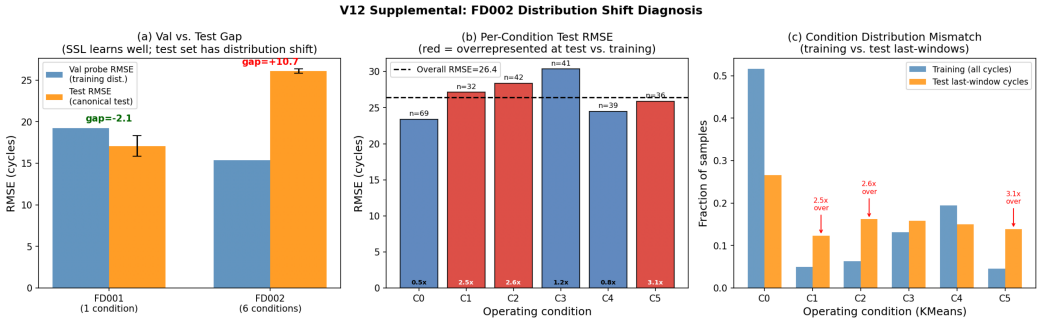

Figure 4: FD002 distribution shift. **(a)** Val vs. test RMSE gap: FD001 has a -2.1 gap (test easier), FD002 has $+10.7$. **(b)** Per-condition test RMSE. **(c)** Condition distributions: conditions 1, 2, 5 severely overrepresented in test.

5 Analysis

5.1 Evaluation Protocol Transparency

The standard C-MAPSS evaluation tests each engine at a single cut point (the last observed cycle). We additionally report sliding-cut RMSE (all cut points from cycle 30 to end), which more honestly measures how the model performs across the full operational range. The sliding-cut RMSE (11.77) is better than last-window (13.98), as expected: at earlier cut points, the encoder has more history and tracks degradation more precisely. We advocate publishing both metrics.

5.2 What Does the Encoder Represent?

PCA analysis. PC1 of the frozen encoder embeddings explains 47.6% of variance with $|\rho| = 0.797$ against the health index. The representation is low-rank and degradation-aligned.

Frozen vs. E2E. The frozen encoder has *higher* tracking ρ (0.856) than E2E (0.804). This is surprising: E2E, which sees RUL labels, tracks degradation *less* faithfully. The explanation is that E2E optimizes for RMSE calibration (mapping to the correct scale) at the expense of rank preservation. The degradation signal originates entirely from pretraining.

Planned: length vs. content. The encoder receives both sensor values and sinusoidal positional encodings indexed by cycle number. Since all training engines are run-to-failure, sequence length correlates with elapsed life. Three planned ablations will disentangle these sources: (1) constant-input test (repeat one cycle's sensors t times, vary t); (2) length-matched cross-engine swap (same t, different sensors); (3) temporal shuffle (permute cycle order within each engine). If the encoder primarily encodes length rather than sensor content, the $R^2 = 0.926$ result would reflect a length artifact and the claims would require revision.

Planned: from-scratch ablation. Training the same architecture with E2E fine-tuning from random initialization (instead of pretrained weights) at each label budget would quantify the pretraining contribution directly. If the delta is <1 RMSE, the E2E result is essentially supervised learning in a Transformer; the paper would then lead exclusively with the frozen and H.I. results as SSL evidence.

6 Limitations

1. **Run-to-failure assumption.** **[TODO: Critical honesty item.]** All C-MAPSS training engines run to failure. The temporal structure of the pretraining data implicitly reveals degradation progression: the encoder sees trajectories that always end at failure, and sequence length correlates with degradation state. This is weaker than labeled RUL supervision, but stronger than truly unlabeled data (e.g., censored industrial data where engines are replaced before failure). The length-vs-content ablation (section 5.2) is required to quantify how

much of the learned signal exploits this structural supervision vs. genuine sensor-content encoding.

2. **Single benchmark.** Results are on C-MAPSS, a simulation with known physics. Generalization to real industrial sensors with noise, missing data, and sensor drift is untested.
3. **Gap to supervised SOTA.** JEPA E2E achieves 14.23 vs. STAR 12.19 ($\sim$2 RMSE gap). Closing this gap likely requires architectural scaling or improved fine-tuning schedules.
4. **FD002/FD004 multi-condition challenge.** The per-condition normalization fails under distribution shift. Condition-aware encoder designs (condition tokens, conditioned normalization) are needed.
5. **Bearing vibration data.** We also evaluated JEPA on single-channel bearing vibration from FEMTO [Nectoux et al., 2012] and XJTU-SY [Wang et al., 2020] (23 run-to-failure episodes, 8 pretraining sources, 33,939 windows). JEPA alone achieved weak correlation with the dominant degradation indicator (spectral centroid PC1 $\rho = 0.071$) and required fusion with handcrafted spectral features to be competitive (Appendix D). This highlights a data-scale limitation: 23 bearing episodes may be insufficient for self-supervised pretraining to discover degradation structure that domain experts encode via spectral centroid, kurtosis, and envelope analysis. The C-MAPSS results (100–260 engines with 14 sensors) suggest the framework becomes effective at larger data scales with richer multivariate structure.

7 Conclusion

We introduced Trajectory JEPA, a self-supervised framework for mechanical prognostics in the grey swan regime. The central finding is that predicting future sensor trajectories in latent space—without any failure labels—causes a Transformer encoder to internalize a latent health variable that is linearly decodable ($R^2 = 0.926$) and tracks within-engine degradation ($\rho = 0.830$). This result is verified against five independent diagnostics and a tight hand-engineered baseline.

The frozen encoder exhibits *higher* tracking fidelity than the E2E fine-tuned version, suggesting that the pretraining objective itself is the source of the degradation signal; fine-tuning refines calibration rather than detection. At low label budgets, this translates to 5–30$\times$ lower variance than supervised approaches—critical for industrial deployment where consistency matters.

The key open question is whether the encoder reads sensor degradation patterns or primarily encodes sequence length. All training data is run-to-failure, providing implicit structural supervision. The planned length-vs-content ablation, from-scratch comparison, and STAR label-efficiency benchmark will determine whether the SSL claims survive scrutiny or need to be scoped to the frozen H.I. recovery result alone. We believe the honest presentation of both the strong results and the open loopholes is itself a contribution: the C-MAPSS community has not previously published within-engine tracking diagnostics, sliding-cut evaluation, or tight feature-regressor baselines alongside last-window RMSE.

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A C-MAPSS Dataset Details

C-MAPSS [Saxena et al., 2008] simulates turbofan engine degradation under controlled conditions. Each engine starts with slightly different initial wear and operates until failure. The simulation produces 21 sensor channels per cycle; we select 14 informative sensors (dropping 7 near-constant channels: s1, s5, s6, s10, s16, s18, s19).

For FD001/FD003 (single operating condition): global min-max normalization per sensor on training data only. For FD002/FD004 (6 operating conditions): per-condition normalization via KMeans clustering ($k = 6$) on operating settings, with min-max applied per cluster. Test data is normalized using training statistics only.

B Hyperparameters

Table 5: Complete hyperparameter settings for Trajectory JEPA.

Component	Hyperparameter	Value
Pretraining	Encoder: d_{model} / layers / heads	256 / 2 / 4
	Encoder parameters	1.26M
	EMA momentum τ	0.99
	Prediction horizon k	$\mathcal{U}[5, 30]$
	Learning rate	3×10^{-4} (cosine decay)
	Epochs / Early stop	200 / probe RMSE at ep. 50
Fine-tuning	Probe	Linear (256 $\rightarrow$ 1)
	Frozen LR / E2E encoder LR	10^{-3} / 10^{-4}
	Epochs / Patience	100 / 20
	Optimizer	AdamW
Data	Train/val split	85% / 15% by engine
	RUL cap R_{max}	125 cycles

C Cross-Fault and Cross-Subset Transfer

Table 6: Cross-fault transfer: FD001-pretrained encoder evaluated on FD003. RMSE in cycles, 5 seeds.

Setting	100%	50%	20%	10%	5%
FD001 in-domain (frozen)	17.81	18.71	19.83	19.93	21.53
FD001$\rightarrow$FD003 (frozen)	$24.79_{\pm 0.6}$	$25.91_{\pm 0.4}$	$27.82_{\pm 1.5}$	$28.79_{\pm 0.4}$	$28.28_{\pm 0.2}$
LSTM (FD001 ref.)	17.36	18.30	18.55	31.22	33.08

Cross-fault transfer incurs 7–9 RMSE penalty, but the cross-fault probe at 10% labels (28.79) still outperforms a supervised LSTM at the same budget (31.22), demonstrating that SSL representations provide useful structure even across mismatched fault modes.

Cross-subset pretraining (FD002$\rightarrow$FD001) matches or slightly outperforms FD001 in-domain frozen at high labels (100%: 17.50 vs 17.81) but underperforms at low labels (10%: 24.45 vs 19.93), indicating that more data from a different distribution does not substitute for matched pretraining data at small scale.

D Bearing Vibration Experiments

To assess whether the framework generalizes to single-channel vibration data at smaller scale, we also evaluated on bearing run-to-failure datasets.

Setup. JEPA pretraining on 33,939 vibration windows from 8 bearing sources (CWRU, MFPT, IMS, FEMTO, XJTU-SY, Paderborn, Ottawa, MAFAULDA), all resampled to 12,800 Hz. The encoder is a 4-layer ViT ($d = 256$, 3.2M parameters) operating on 16 patches of 64 samples each, with 62.5% masking. Evaluation on FEMTO [Nectoux et al., 2012] and XJTU-SY [Wang et al., 2020] (23 episodes, 75%/25% split).

Results. JEPA alone (RMSE 0.189) matches handcrafted features (HC+LSTM: 0.177, $p = 0.40$) but has very low correlation with the dominant degradation indicator (spectral centroid PC1 $\rho = 0.071$). A hybrid architecture fusing JEPA embeddings with 18 handcrafted spectral features achieves RMSE 0.055 ± 0.004 ($+75.5\%$ vs. time-only baseline), confirming that JEPA captures complementary information. For cross-dataset transfer (FEMTO→XJTU-SY), temporal contrastive pretraining outperforms JEPA by $+18.8\%$ ($p < 0.001$).

Lesson. The bearing results highlight a data-scale limitation. With 23 run-to-failure episodes and single-channel vibration, JEPA pretraining does not discover spectral centroid shift—the dominant degradation indicator—that domain experts encode directly. The framework requires either (a) larger datasets with richer structure (as in C-MAPSS) or (b) domain-informed masking strategies that direct the encoder toward degradation-relevant features.

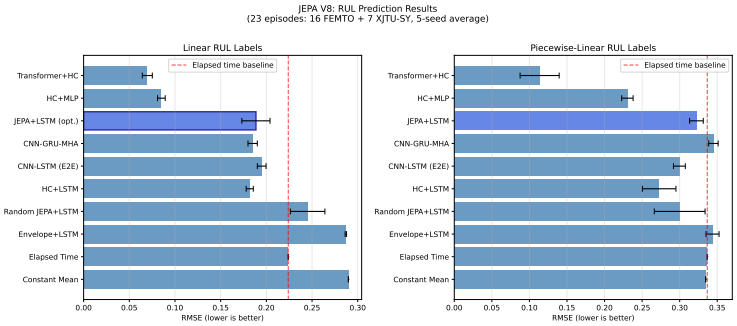

Figure 5: Bearing RUL comparison (FEMTO + XJTU-SY, 5 seeds). The hybrid JEPA + HC architecture achieves the best result, but JEPA alone is only marginally better than handcrafted features—and much worse than the hybrid. This motivates the trajectory formulation on richer multivariate data (C-MAPSS).